GoPAL-AI Master Blueprint
A living sanctuary for language, memory, companionship and discovery
Repository: [jamlove1318-lab/GoPAL-AI]https://github.com/jamlove1318-lab/GoPAL-AI
Blueprint date: 12 October 2026

1. North star - a place to return to

GoPAL-AI should feel less like a checklist app and more like a living place where a learner is known, welcomed and gently invited back.

"My world continued. Cassidy remembers what matters. I can explore and practice without being judged by a streak."

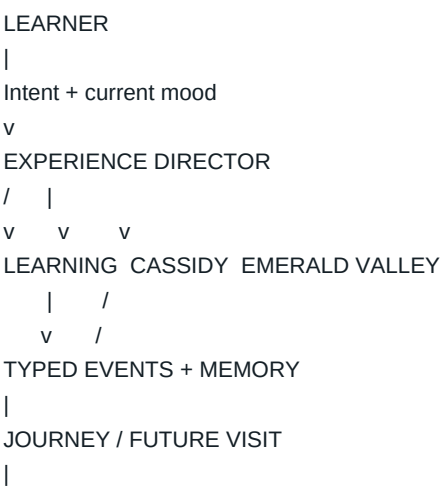
Learning is the purpose; world, character, story and atmosphere make practice meaningful.

Design principles

- Continuity over resets.
- Curiosity over obligation.
- Relationship over score.
- Context over disconnected drills.
- Small meaningful surprises over noisy notifications.
- Agency over rigid paths.
- Honest memory over simulated intimacy.

2. Illustrated experience model

text



The learner should not need to navigate this architecture. Every interaction should answer: Where am I? Why is this happening now? What can I do next? Use one primary action, a few meaningful secondary choices and optional detail.

3. Emotional rhythm and re-entry

1. Arrival: time-aware greeting, subtle environmental change and a concise "while you were

away" recap only when something meaningful happened.

2. Invitation: offer options suited to intent and energy-talk, study, explore, create,

revisit a memory or simply spend a moment in the sanctuary.

3. Engagement: language practice emerges naturally from the chosen activity. Feedback is

specific, kind and actionable.

4. Consequence: Cassidy, a resident, a plant or a location remembers the outcome and changes

in a coherent way.

5. Departure: save context and offer a soft stopping point. Recover unfinished activities

without guilt or punishment for absence.

The app should never make a learner feel they failed by being away.

4. Visual direction - Emerald Valley

Mood: a warm, cinematic illustrated sanctuary with emerald greens, jade accents, sunlit wood, paper, stone, water, lantern light and seasonal color. It should feel tactile, inhabited and slightly mysterious.

Environmental storytelling

- Cool dawn, warm morning, golden late afternoon and lantern-lit evening.
- Rain-softened light, seasonal shifts and festival accents.
- Layered foreground, midground and background; readable silhouettes and gentle depth.
- Subtle movement and clear touch targets, not constant particles or a noisy game HUD.
- Readable UI typography; expressive display typography reserved for chapters and landmarks.
- Respect contrast, scalable text, localization and reduced-motion preferences.

Audio: location-specific ambience and optional music layers, independent volume controls, ducking, mute, sleep-friendly defaults and offline fallback. Never surprise the learner with loud playback.

5. Learning - meaningful language in context

Adaptive conversation

Infer level, intent, recent errors and confidence; choose a scenario where language matters. Learners control hints, difficulty and correction intensity.

Socratic roleplay

Ask questions that help learners think and respond instead of always revealing the answer first. Offer a model answer or translation when requested.

Residents as learning partners

Residents have roles, routines, preferences and continuity. They are people in the world, not merely decorative chatbot wrappers.

Pronunciation and listening

Pitch/prosody step-through, shadowing, replay, slower audio, syllable highlighting and comparison. Be transparent about recognition confidence and never overstate automated pronunciation accuracy.

Writing and calligraphy

Stroke-order guidance, brush/canvas practice and saved creations. Explain language patterns and allow retries rather than simply marking answers wrong.

Vocabulary and spaced recall

Capture useful words from scenes, attach context and difficulty, schedule revisits and measure retention over time-not raw taps.

Evaluation and Journey

Combine comprehension, productive use, recall, fluency and self-reported confidence.

Separate formative feedback from a single opaque score. Save dialogue with script, phonetics/romaji where useful, translation and replayable audio drills. Show "then vs now" only with learner consent.

6. Cassidy - a companion, not a meter

Cassidy should have a consistent voice, boundaries, preferences, life rhythm and a relationship built through shared experiences.

Conceptual layers

- Identity and personality.
- Perception of learner and world context.
- Action and bounded autonomy.
- Interaction lifecycle.
- Learning reactions.
- World presence.
- Presentation and renderer.

Give each layer one documented responsibility with explicit inputs and outputs.

Memory model: separate session context, learner preferences, shared experiences, milestones and world facts. Save only useful information, allow correction/deletion, and never pretend to remember an event that is absent from memory.

Relationship expression: a remembered joke, a letter, an invitation, a changed greeting, a festival memory or a companion plant-not a visible affection score.

Autonomy: low-pressure moments, respect for user intent, quiet mode, privacy and notification limits. Never create guilt for missed activity.

Presentation: one canonical identity/asset manifest, robust 2D fallback, consistent scale/lighting/animation, graceful loading/error states and validated renderer contracts. A production asset is not proof that runtime 3D is wired.

7. Emerald Valley - a world with continuity

Locations should have a distinct sensory identity, purpose, inhabitants, discoverable details and evolving state.

World state dimensions: day cycle, weather, season, festival, location familiarity, resident schedules, discovered lore, object state, activity outcomes and learner-created artifacts.

text

```
CLOCK + ENVIRONMENT + LEARNER HISTORY
|
v
EVENT DIRECTOR
|
SCENE / RESIDENT / ACTIVITY
|
LANGUAGE OPPORTUNITY
|
LEARNER ACTION
|
EVENT BUS + PERSISTENCE
```

|
MEMORY + WORLD STATE CHANGE
|
NEXT VISIT FEELS ALIVE
Objects should matter beyond collecting: a plant grows through care, a wish plaque preserves an intention, a landmark reveals new context, a resident changes a routine, and a familiar path gains details.
Discovery should be layered, not a grind. Offline state should remain coherent: local state and downloaded assets usable, queued sync safe, conflict rules deterministic, and learner creations never silently lost.
8. Activities in the living sanctuary
Activity / surface	Experience intent
Home / Sanctuary	Greeting, environmental scene, continuity cards, relevant invitations and easy access to Study, World and Journey
Study	Roleplay, contextual drills, pitch/prosody, calligraphy, vocabulary recall, listening, transcript archive and creative tasks
World Map	Locations, resident profiles, familiarity, atmospheric transitions, lore inspection and clear discovered/accessible states
Living Bonsai	Care choices and visible growth; absences should not kill it or create guilt
Matcha workshop	Sequence, measurement, vocabulary and cultural context, with accessible alternatives to timed gestures
Soundscape / World Radio	Separate volumes, mute, optional ambience and sleep-friendly defaults
Ema wish ritual	Private intention with visibility control and optional reminders
Quests / souvenir shop	Optional narrative goals, seeds, fertilizer and artifacts that reward expression/discovery, not grinding
Creation Studio / Museum	Save/reopen learner work, postcards, cultural keepsakes and story memories with attribution and privacy controls
Journey	Personal timeline, dialogue, milestones and capability growth-not leaderboards
Settings / diagnostics	Memory/privacy, accessibility, storage health, asset/runtime checks and understandable errors

Each activity must teach or express something, produce a meaningful consequence, and save/reopen reliably.
9. How systems work together
1. Presentation layer: screens, accessible components, Cassidy renderer and world scene. UI

must not duplicate domain rules.
2. Experience Director: turns intent and context into a small set of useful choices and

explains why an activity is suggested.
3. Domain engines: learning/tutor, Cassidy, world/environment, Journey/memory, audio,

creation, knowledge and quest/economy. Each owns a narrow responsibility.
4. Event bus: typed events such as LessonCompleted, ResidentConversationFinished,

ObjectDiscovered, PlantCaredFor, MemorySaved and LocationVisited.

5. Memory/state services: canonical schemas, consent boundaries, persistence adapters,

migrations, offline queue and sync-conflict policy.

6. Observability: trace user action -> director -> engine -> event -> storage -> rendered

consequence. Redact private dialogue and secrets.

Core product loop: intent -> activity -> language use -> feedback -> meaningful outcome -> memory/world change -> later recall and re-entry.

10. Privacy, accessibility and trust

- Explain what is remembered, why it helps, and how to edit/delete/export it.
- Clarify local versus synced data and apply data minimization.
- Keep secrets server-side; validate Supabase row-level security. Never ship service-role keys in the mobile app.
- Support screen readers, sufficient contrast, scalable text, reduced motion, captions/transcripts, adjustable audio, adequate touch targets and alternatives to timed gestures.
- Provide quiet mode and optional reminders. Avoid streak shame, manipulative scarcity, forced comparison and deceptive AI-memory claims.
- Avoid raw private conversation logs in diagnostics; document retention and opt-in analytics where appropriate.

11. Roadmap and acceptance gates

Phase 1 - Reliable foundation

Route inventory, build/typecheck, navigation smoke tests, state ownership, persistence contracts, error boundaries and offline restart.

Phase 2 - One complete learning loop

Intent -> contextual roleplay -> feedback -> transcript/vocabulary -> later recall -> visible growth in Journey.

Phase 3 - Cassidy continuity

Canonical assets, tested memory boundaries, contextual reactions, predictable autonomy and 2D fallback.

Phase 4 - Emerald Valley continuity

Environment clock, event director, resident routines, object state, activity outcomes and return-visit changes.

Phase 5 - Activities and archive

Bonsai, matcha, calligraphy, audio, wishes, quests, creation and Museum; each saves/reopens and offers accessible alternatives.

Phase 6 - Trust and polish

Performance budgets, asset size, battery/audio policy, localization, privacy controls, telemetry and regression suite.

Acceptance gate for every feature: reachable route; successful happy path; loading/error/empty states; persisted outcome; restart/re-entry behavior; accessibility; offline policy; automated test; documented owner.

12. Preservation rule and final principle

Do not delete ambiguous legacy systems. Trace references, archive or deprecate additively,

compare behavior, and get explicit agreement before removal.
GoPAL-AI should not ask the learner to maintain an app. It should invite them back into a place they care about.
Success is meaningful language use, remembered context, a world that responds, a companion who feels consistent and a learner who notices their own growth. Build fewer disconnected features and more complete loops.